

FTML practical session 12

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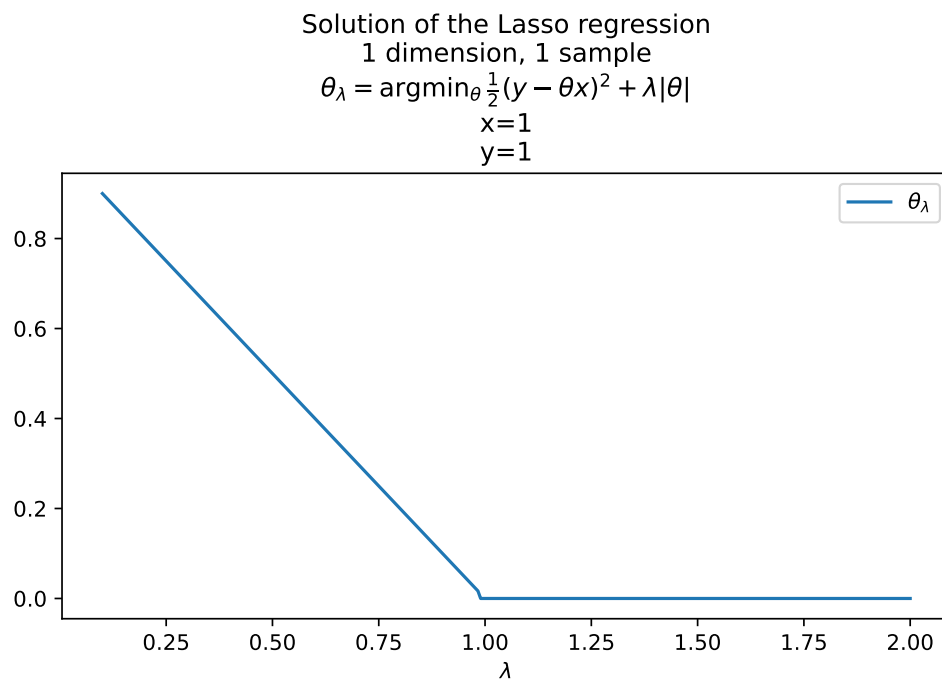


FIGURE 1 – Solution of the one-dimensional Lasso estimator

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1 GENERAL LASSO ESTIMATOR

Lasso regression is a regularization method for linear regression based on the L1 norm. With our usual notations, the Lasso estimator is the solution to the following optimization problem :

$$\tilde{\theta}_\lambda \in \arg \min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{2} \|y - X\theta\|^2 + \lambda \|\theta\|_1 \right\} \quad (1)$$

where

- $X \in \mathbb{R}^{n,d}$ is the design matrix.
- $y \in \mathbb{R}^n$ is the vector of labels
- $\lambda \in \mathbb{R}_+$ is the regularization constant (called alpha in scikit-learn)

2 1D LASSO

In this session we analyze the Lasso regression on a simple, one-dimensional example, in order to develop the intuition of why this method leads to **sparse** estimators. We hence consider the simplified problem :

$$\theta_\lambda = \arg \min_{\theta} F_\lambda(\theta) \quad (2)$$

with

$$F_\lambda(\theta) = \frac{1}{2} (y - \theta)^2 + \lambda |\theta| \quad (3)$$

and $y \in \mathbb{R}_+$, $\lambda \in \mathbb{R}_+$, $\theta \in \mathbb{R}$, where we assume that the input x is equal to 1 in order to lighten the notations.

Find the solution to the one-dimensional Lasso problem, as a function of λ . An example solution for $y = 1$ is shown in figure 1.

Remark : in order to conclude on the optimal θ value as a function of λ , you can set an arbitrary value of y .

Run a simulation that reproduces the plot shown in figure 1 using scikit-learn.